問題 272

次の三角形で α の三角比を求めよ.

解答:

(1) 三辺 $\sqrt{2}, \sqrt{7}, 3$（斜辺 3）

$\sin\alpha = \dfrac{\sqrt{2}}{3}$, $\cos\alpha = \dfrac{\sqrt{7}}{3}$, $\tan\alpha = \dfrac{\sqrt{2}}{\sqrt{7}} = \dfrac{\sqrt{14}}{7}$

(2) 三辺 $1, 2, \sqrt{5}$（斜辺 $\sqrt{5}$）

$\sin\alpha = \dfrac{2}{\sqrt{5}} = \dfrac{2\sqrt{5}}{5}$, $\cos\alpha = \dfrac{1}{\sqrt{5}} = \dfrac{\sqrt{5}}{5}$, $\tan\alpha = 2$

問題 273

図において，$\tan 15^\circ$ の値を求めよ.

解答:

加法定理：$\tan 15^\circ = \tan(45^\circ - 30^\circ) = \dfrac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} = \dfrac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \dfrac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3}$

問題 274

傾斜角 34°，距離 815 m のケーブルカーの標高差.

解答:

$815 \sin 34^\circ = 815 \times 0.5592 \approx 456\,\mathrm{m}$

問題 275

次の式の値を求めよ.

解答:

(1) $\sin 60^\circ \cos 30^\circ + \cos 120^\circ \sin 150^\circ + \sin 135^\circ \cos 180^\circ$

$= \dfrac{3}{4} - \dfrac{1}{4} - \dfrac{\sqrt{2}}{2} = \dfrac{1-\sqrt{2}}{2}$

(2) $\dfrac{\tan 30^\circ - \tan 135^\circ - \tan 180^\circ}{1 + \tan 120^\circ \tan 45^\circ}$

$= \dfrac{\frac{1}{\sqrt{3}} + 1 - 0}{1 - \sqrt{3}} = \dfrac{\frac{\sqrt{3}+1}{\sqrt{3}}}{1-\sqrt{3}} = \dfrac{\sqrt{3}+1}{\sqrt{3}(1-\sqrt{3})} = -\dfrac{2\sqrt{3}+3}{3}$

問題 276

α 鈍角，$\cos\alpha = -\dfrac{2}{5}$ のとき.

解答:

$\sin\alpha = \dfrac{\sqrt{21}}{5}$, $\tan\alpha = -\dfrac{\sqrt{21}}{2}$

問題 277

α 鈍角，$\tan\alpha = -3$ のとき.

解答:

$\cos\alpha = -\dfrac{\sqrt{10}}{10}$, $\sin\alpha = \dfrac{3\sqrt{10}}{10}$

問題 278

△ABC の計算.

解答:

(1) $a = \sqrt{6}$, $B = 105^\circ$, $C = 30^\circ \to A = 45^\circ$

$2R = \dfrac{\sqrt{6}}{\sin 45^\circ} = 2\sqrt{3}$, $R = \sqrt{3}$, $c = 2\sqrt{3}\sin 30^\circ = \sqrt{3}$

(2) $a = 3$, $b = 5$, $C = 120^\circ$

$c^2 = 9 + 25 + 15 = 49$, $c = 7$, $S = \dfrac{15\sqrt{3}}{4}$

(3) $a = 3$, $b = 8$, $c = 7$

$\cos B = \dfrac{9 + 49 - 64}{42} = -\dfrac{1}{7}$, $\sin B = \dfrac{4\sqrt{3}}{7}$, $S = 6\sqrt{3}$